JACK CASE

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EDUCATION

Doctor of Philosophy in Cell and Systems Biology — Incoming Fall 2026
University of Toronto (Scarbrough), Department of Cell and Systems Biology
Supervisor Dr. Christina Guzzo

Master of Microbiology & Immunology
Dalhousie University, Department of Microbiology & Immunology — 2026
Thesis: Fitness costs associated with baloxavir marboxil resistance mutations in influenza A virus
Supervisor Dr. Denys Khaperskyy

Honours Bachelor of Science — 2024
Certificate in Genetics
Dalhousie University, Department of Microbiology & Immunology

RESEARCH EXPERIENCE

Master's Student
Department of Microbiology & Immunology
Supervisor: Dr. Denys Khaperskyy

- Investigated the effect of antiviral resistance mutations on influenza A virus host shutoff.
- Used cell-based infection and transfection assays to assess viral fitness & protein function.
- Presented my research at multiple local, national, & international research events.
- This work culminated in a first author paper published in PLOS Pathogens.

Honours Student
Department of Microbiology & Immunology
Supervisor: Dr. Denys Khaperskyy

- Conducted basic exploratory research onto the effect of antiviral resistance mutations on the influenza A virus protein PA-X using transfection-based cell assays.
- Presented my research at local research events, as well as at a national conference.

PUBLICATIONS

Case JR, Khaperskyy DA (2026) Impaired host shutoff is a fitness cost associated with baloxavir marboxil resistance mutations in influenza A virus PA/PA-X nuclease domain. PLOS Pathogens 22(2): e1013550. https://doi.org/10.1371/journal.ppat.1013550

Presentations

- Impaired host shutoff is a fitness cost associated with baloxavir marboxil resistance mutations in influenza A virus PA/PA-X nuclease domain. **Zoonoses, AMR, and BioInformatics (ZAMBI) Meeting**, July 7, 2026. Dalhousie University, Halifax, Nova Scotia. Oral Presentation.
- Impaired host shutoff is a fitness cost associated with baloxavir marboxil resistance mutations in influenza A virus PA/PA-X nuclease domain. **Canadian Society for Virology 2026 Symposium**, June 3, 2026. Guelph University, Guelph, Ontario. Oral Presentation.
- Effects of baloxavir resistance mutations on PA-X mediated host shutoff by influenza A virus. **Van Rooyen Symposium**, October 24, 2025. Dalhousie University, Halifax, Nova Scotia. Poster Presentation.
- Effects of baloxavir resistance mutations on PA-X mediated host shutoff by influenza A virus. **44th Meeting of the American Society for Virology**, July 14-18, 2025. Palais des congrès de Montreal, Montreal, Quebec. Oral Presentation.
- Effects of baloxavir resistance mutations on PA-X mediated host shutoff by influenza A virus. **Canadian Centre for Vaccinology Research Day**, April 15, 2025. Lord Nelson Hotel, Halifax, Nova Scotia. Poster Presentation.
- Effects of baloxavir resistance mutations on PA-X mediated host shutoff by influenza A virus. **Canadian Society for Virology 2024 Symposium**, June 2-5, 2024. University of Saskatchewan Campus, Saskatoon, Saskatchewan. Poster Presentation.
- Influenza A virus mutations that confer baloxavir acid resistance negatively affect the host shutoff endonuclease PA-X. **Canadian Centre for Vaccinology Research Day**, April 23, 2024. Lord Nelson Hotel, Halifax, Nova Scotia. Oral Presentation.

AWARDS

- Dalhousie University Faculty of Medicine Excellence in Research Award 2026 – Masters ($500)
- Canadian Society for Virology Research Excellence Award 2026 - Masters ($500)
- Nova Scotia Graduate Scholarship – Master's ($10,000/year, 2 years)
- Van Rooyen Symposium – Best Poster Presentation ($100)
- 44th Meeting of the American Society for Virology – Travel Award ($~800)
- Canadian Society for Virology 2024 and 2026 Symposium – Travel Award (~$800)
- Department of Microbiology & Immunology Easterbrook Travel Award ($1000)
- Canadian Centre for Vaccinology Research Day 2025 – Best Poster Presentation ($300)

TEACHING EXPERIENCE

Teaching Assistant, MICI3114 Virology, Dalhousie University, Fall 2024 & 2025

- Ran biweekly tutorials for 40 students, designing review questions, short quizzes, and teaching core concepts covered in class.
- Graded assignments and exams.

Teaching Assistant, MICI2400 Laboratory Methods in Microbiology and Immunology, Dalhousie University, Winter 2025 & 2026

- Lab-based course where I directly supervised ~15 students as they performed laboratory exercises, answering questions, assessing performance, and fostering a supportive environment. I also assisted with laboratory preparation and clean up.
- Graded assignments and practical exams.

PROFESSIONAL MEMBERSHIPS

- Canadian Society for Virology 2024-present
- American Society for Virology 2025-present